LESSON 01 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO-flavoured)

Beyond the core syllabus sentence (principal values), the analysis of composite inverse-trig functions — piecewise closed forms, parity/period, graph shapes, extrema on intervals — is standard olympiad and coaching-extension material. Every model here is ***derivable from Part I*** *(branch reads + addition formulas): no new theory is assumed, only sustained bookkeeping. Nothing here "can never appear" in JEE Advanced; it is simply less typical than the Part I types.*

The full closed form of f(x) = sin⁻¹(sin x) + cos⁻¹(cos x)

Both pieces are sheet-02 reads. Reduce x *to* $t \in [0, 2\pi)$ *first (both components have period* 2π*, so* f *has period* 2π*). On* $[0, 2\pi)$*:*

$$\sin^{-1}(\sin t) = t \ (0 \le t \le \tfrac{\pi}{2}), \qquad = \pi - t \ (\tfrac{\pi}{2} < t < \tfrac{3\pi}{2}),$$

$$= t - 2\pi \ (\tfrac{3\pi}{2} \le t < 2\pi)$$

$$\cos^{-1}(\cos t) = t \ (0 \le t \le \pi), \qquad = 2\pi - t \ (\pi < t < 2\pi)$$

Adding on each sub-interval of $[0, 2\pi)$ *— leg by leg (the composition of two affine pieces is affine, so only endpoints need checking):*

- $t \in [0, \tfrac{\pi}{2}]$: $t + t = 2t$ *(rises* $0 \to \pi$*).*
- $t \in [\tfrac{\pi}{2}, \pi]$: $(\pi - t) + t = \pi$ *(the plateau).*
- $t \in [\pi, \tfrac{3\pi}{2}]$: $(\pi - t) + (2\pi - t) = 3\pi - 2t$ *(falls* $\pi \to 0$*).*
- $t \in [\tfrac{3\pi}{2}, 2\pi)$: $(t - 2\pi) + (2\pi - t) = 0$ *(the floor).*

$$f(t) = 2t \text{ on } [0, \tfrac{\pi}{2}], \qquad f(t) = \pi \text{ on } [\tfrac{\pi}{2}, \pi]$$

$$f(t) = 3\pi - 2t \text{ on } [\pi, \tfrac{3\pi}{2}], \qquad f(t) = 0 \text{ on } [\tfrac{3\pi}{2}, 2\pi)$$

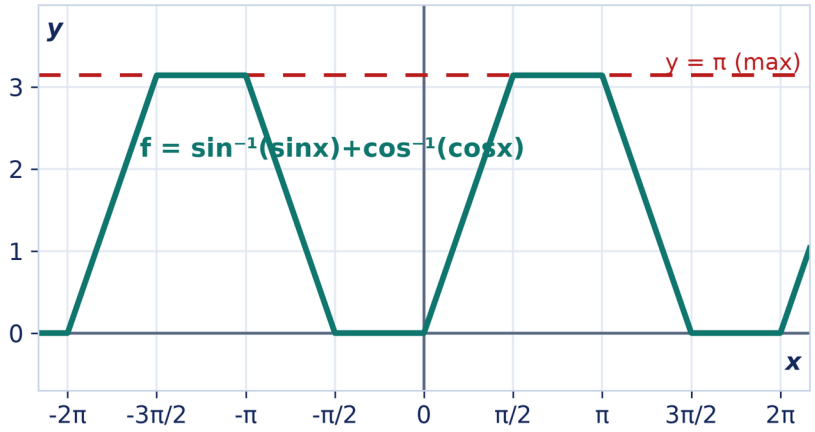
Range on $\mathbb{R}$*:* f *takes every value in* $[0, \pi]$ *(max* π *on the plateau, min* 0*).* ***Neither odd nor even****:* $f(x + 2\pi) = f(x)$*.*

SHORT DERIVATION

The bookkeeping lesson — always split at the branch knots

The knots of $\sin^{-1}(\sin x)$ *are* $x = \frac{\pi}{2} + k\pi$ *(where the sawtooth turns); those of* $\cos^{-1}(\cos x)$ *are* $x = k\pi$*. A sum's piecewise form splits at the* ***union*** *of knots. On each open piece both components are linear* $(\pm x + \text{const})$*, so* f *is affine there: the whole problem reduces to assembling slopes. Slopes of* $\sin^{-1}(\sin x)$ *alternate* $+1, -1$*; of* $\cos^{-1}(\cos x)$*:* $+1, -1$*; sums have slopes in* $\{-2, 0, 2\}$ *— the flat middle leg above is the* $+1 + (-1)$ *collision.*

f(x): plateau π on [π/2, π], zero on [3π/2, 2π), period 2π — range [0, π]



Sibling composites worth owning

- $g(x) = \sin^{-1}(\sin x)$ *alone: odd,* 2π*-periodic, range* $[-\pi/2, \pi/2]$*.*
- $h(x) = \cos^{-1}(\cos x)$*: even,* 2π*-periodic, range* $[0, \pi]$ *— note* $h(x) = |x|$ *on* $[-\pi, \pi]$*, the "triangle wave".*
- $p(x) = \tan^{-1}(\tan x)$*: odd,* π*-periodic (undefined at odd multiples of* $\pi/2$*), range* $(-\pi/2, \pi/2)$ *— the sawtooth of sheet 02.*
- $q(x) = \sin^{-1}(\sin x) + \cos^{-1}(\sin x)$*: collapses to* $\frac{\pi}{2}$ ***everywhere*** *(complementary pair with the same argument* $\sin x$ *— a one-liner that looks like work).*
- $r(x) = \tan^{-1}(\sin x)$*: odd,* 2π*-periodic, range* $[-\frac{\pi}{4}, \frac{\pi}{4}]$ *(bound* $\sin x \in [-1, 1]$ *through the increasing* $\tan^{-1}$*). Extrema-type questions ("range of* $\tan^{-1}(\sin x) + \tan^{-1}(\cos x)$ *") are solved by monotone-plus-derivative-free bounding:* $\sin x + \cos x$*-style collapse.*

STANDARD APPLICATION

Extrema method for $u(x) = \tan^{-1}(\sin x) + \tan^{-1}(\cos x)$

Add via the table: $\tan^{-1} u + \tan^{-1} v = \tan^{-1} \frac{u+v}{1-uv}$ when $uv < 1$ — here $uv = \sin x \cos x \le \frac{1}{2} < 1$ ✓ always. So

$$u(x) = \tan^{-1}\left(\frac{\sin x + \cos x}{1 - \sin x \cos x}\right)$$

a monotone transform of $s = \sin x + \cos x \in [-\sqrt{2}, \sqrt{2}]$ with $\sin x \cos x = \frac{s^2-1}{2}$. Write $u = \tan^{-1} \frac{s}{1-\frac{s^2-1}{2}} = \tan^{-1} \frac{2s}{3-s^2}$. On $s \in [-\sqrt{2}, \sqrt{2}]$ the fraction $\frac{2s}{3-s^2}$ is odd-increasing in s (denominator $3 - s^2 \in [1, 3]$ stays positive), so max at $s = \sqrt{2}$: $\frac{2\sqrt{2}}{1} = 2\sqrt{2}$, i.e. $\max u = \tan^{-1}(2\sqrt{2})$ and $\min u = -\tan^{-1}(2\sqrt{2})$; range $\left[-\tan^{-1} 2\sqrt{2},\ \tan^{-1} 2\sqrt{2}\right]$. Every step used only Part I: the addition formula's condition, the range of $\sin x + \cos x$, monotonicity of $\tan^{-1}$.

COMMON TRAP

Traps specific to piecewise work

- Splitting at the wrong knots (only one component's) creates false affine pieces with wrong slopes.
- Period: $\sin^{-1}(\sin x) + \cos^{-1}(\cos x)$ has period 2π; $\tan^{-1}(\tan x)$ has π — sums take the LCM (2π here).
- Endpoints of branch windows ($x = \frac{\pi}{2}$ etc.) are continuous for these composites but not differentiable — extrema questions may land on a knot; evaluate both one-sided values (they agree here).
- In "$q(x)$ collapses to $\pi/2$" the argument of both inverses must be identical; $\sin^{-1}(\sin x) + \cos^{-1}(\cos x) \neq \frac{\pi}{2}$ (that is f above!).

LESSON 02 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO standard)

Telescoping sums of $\tan^{-1}$-terms are a staple of IOQM/RMO arithmetic. The engine is Part I's difference formula; the skill is recognising which $\frac{a-b}{1+ab}$ is hiding inside the term. Sums are treated exactly (finite forms and the convergent infinite limits); no calculus beyond "$\tan^{-1} n \to \frac{\pi}{2}$ as $n \to \infty$" is used (that limit is the graph of $\tan^{-1}$ from Part I).

The one identity every such sum hides

KEY RESULT

Difference identity (the telescoping workhorse)

For $ab > -1$:

$$\tan^{-1} a - \tan^{-1} b = \tan^{-1}\left(\frac{a-b}{1+ab}\right)$$

So any term $\tan^{-1}\frac{c}{d}$ is telescoping material the moment $c = a-b$ and $d = 1+ab$ for an integer-valued sequence a_k, b_k (usually $b_k = a_{k+1}$ or $a_k = k + \text{const}$). With $a = k+1$, $b = k$: $\frac{a-b}{1+ab} = \frac{1}{1+k+k^2}$; with $a = k+1$, $b = k-1$: $\frac{2}{k^2}$. Those two shapes cover most published problems.

SHORT DERIVATION

Series I — $\displaystyle\sum_{k=1}^{n} \tan^{-1}\frac{1}{1+k+k^2} = \tan^{-1}(n+1) - \frac{\pi}{4}$

Each term: $\frac{1}{1+k(k+1)} = \frac{(k+1)-k}{1+k(k+1)} = \tan^{-1}(k+1) - \tan^{-1} k$ (valid: $k(k+1) > -1$ ✓).

The sum telescopes:

$$\sum_{k=1}^{n} \left[\tan^{-1}(k+1) - \tan^{-1} k\right]$$

$$\tan^{-1}(n+1) - \tan^{-1} 1 = \tan^{-1}(n+1) - \frac{\pi}{4}.$$

Checks: $n = 1$: LHS $= \tan^{-1}\frac{1}{3} \approx 0.3217$; RHS $= \tan^{-1} 2 - \frac{\pi}{4} \approx 1.1071 - 0.7854 = 0.3217$ ✓. As $n \to \infty$: sum $\to \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$.

SHORT DERIVATION

Series 2 — $\sum_{k=1}^{n} \tan^{-1}\frac{2}{k^2} = \tan^{-1}(n+1) + \tan^{-1} n - \frac{\pi}{4}$

Here $\frac{2}{k^2} = \frac{(k+1)-(k-1)}{1+(k+1)(k-1)}$, so $\tan^{-1}\frac{2}{k^2} = \tan^{-1}(k+1) - \tan^{-1}(k-1)$ (valid since $(k+1)(k-1) = k^2 - 1 > -1$). This is a **two-step** telescope (stride 2):

$$\sum_{k=1}^{n} [\tan^{-1}(k+1) - \tan^{-1}(k-1)]$$

$$[\tan^{-1}(n+1) + \tan^{-1} n] - [\tan^{-1} 0 + \tan^{-1} 1]$$

$$= \tan^{-1}(n+1) + \tan^{-1} n - \frac{\pi}{4}.$$

Checks: $n = 1$: $LHS = \tan^{-1} 2 \approx 1.1071$; $RHS = \tan^{-1} 2 + \tan^{-1} 1 - \frac{\pi}{4} = 1.1071$ ✓.

Infinite limit: $\frac{\pi}{2} + \frac{\pi}{2} - \frac{\pi}{4} = \frac{3\pi}{4}$ — the classic "$\sum_{k=1}^{\infty} \tan^{-1}\frac{2}{k^2} = \frac{3\pi}{4}$".

Worked variants — the same engine, different residues

WORKED EXAMPLE

Variant A — $\sum_{k=1}^{n} \tan^{-1}\frac{2k}{k^4+k^2+2}$ **(the denominator is the tell)**

Spot $a = k^2 + k + 1$, $b = k^2 - k + 1$: then $a - b = 2k$ ✓ and, from $(k^2+1)^2 - k^2 = k^4 + k^2 + 1$,

$$1 + ab = 1 + (k^2 + k + 1)(k^2 - k + 1)$$

$$= k^4 + k^2 + 2$$

exactly the given denominator. Each term is $\tan^{-1}(k^2 + k + 1) - \tan^{-1}(k^2 - k + 1)$, and $k^2 - k + 1 = (k-1)^2 + (k-1) + 1 = u_{k-1}$ with $u_k = k^2 + k + 1$ — the sum telescopes along u_k:

$$\sum_{k=1}^{n} \tan^{-1}\frac{2k}{k^4+k^2+2}$$

$$= \tan^{-1}(n^2 + n + 1) - \tan^{-1} 1$$

$$= \tan^{-1}(n^2 + n + 1) - \frac{\pi}{4}.$$

Numeric verification (also in `math_checks.py`): $n = 1$: $\tan^{-1}\frac{2}{4} = 0.46365 = \tan^{-1} 3 - \frac{\pi}{4}$ ✓. $n = 2$: $\tan^{-1} 0.5 + \tan^{-1}\frac{2}{11} = 0.64350 = \tan^{-1} 7 - \frac{\pi}{4}$ ✓.

STANDARD APPLICATION

Strategy card for arctan series

1. Force the term into the shape $\tan^{-1}\frac{a-b}{1+ab}$ by **matching the denominator to** $1+ab$ with small integer a_k, b_k.
2. Ensure $a_k b_k > -1$ on the range (automatic for positive sequences).
3. Telescoping: write out the first two and last two terms before cancelling; stride-2 series lose half their terms to the boundary.
4. Infinite sums: take $n \to \infty$ in the closed form, using $\tan^{-1} n \to \frac{\pi}{2}$; never sum "infinitely many small arctans" by estimation.
5. Verify $n = 1$ and $n = 2$ numerically before trusting the closed form.

COMMON TRAP

Traps — series

- Stride-2 telescopes drop the parity of surviving boundary terms; write n even and $n = 5$ by hand and compare.
- $\tan^{-1}\frac{2}{k^2}$ at $k = 1$: $\tan^{-1} 2$; the split $\tan^{-1} 2 - \tan^{-1} 0$ is legal ($0 \cdot 2 = 0 > -1$ ✓) — but at $ab = -1$ the difference formula is void: e.g. $a = 1, b = -1$ gives $\tan^{-1} 1 - \tan^{-1}(-1) = \frac{\pi}{2}$ while $\frac{a-b}{1+ab} = \frac{2}{0}$ explodes — the boundary case is real, not a formality.
- Sign of the residue: $\tan^{-1}\frac{1}{1+k+k^2}$ telescopes as $\tan^{-1}(k+1) - \tan^{-1} k$ (positive), not $k - (k+1)$.

LESSON 03 / 03

ASSUMPTIONS & CONVENTIONS

STATUS — olympiad/specialised extension (IOQM/RMO + coaching classics)
These are "prove/evaluate with structure" problems: symmetric conditions on x, y, z forcing angle sums, nested-inverse expressions, and range problems where the branch constraints are the hidden system of inequalities. All are derivable from Part I; the added value is problem-recognition. No claim is made that such items "can never appear" in JEE Advanced — they are simply the harder end of the same toolkit.

The classic: x + y + z = xyz forces A + B + C = π

KEY RESULT

Theorem (with its hypothesis)
If $x, y, z > 0$ and $x + y + z = xyz$, then

$$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi.$$

More symmetrically: with $A = \tan^{-1} x$ etc. all in $(0, \frac{\pi}{2})$, the condition $x + y + z = xyz$ is exactly $\tan(A + B + C) = 0$ — and the range bookkeeping ($A + B + C \in (0, \frac{3\pi}{2})$) pins $A + B + C = \pi$ (it cannot be 0: each angle is positive).

SHORT DERIVATION

Proof — expand $\tan(A + B + C)$
With $A = \tan^{-1} x, B = \tan^{-1} y, C = \tan^{-1} z$:

$$\tan(A + B + C) = \frac{x + y + z - xyz}{1 - (xy + yz + zx)}.$$

Hypothesis: the numerator vanishes. So either the denominator is nonzero and $\tan(A + B + C) = 0$, giving $A + B + C = k\pi$; since $A + B + C \in (0, \frac{3\pi}{2})$ the only candidate is π. Or the denominator vanishes too ($xy + yz + zx = 1$): then $A + B + C \in \{\frac{\pi}{2}, \frac{3\pi}{2}\}$ — one checks directly that $x, y, z > 0$ with $x + y + z = xyz$ forces $xy + yz + zx > 1$, so this degenerate case cannot occur (e.g. by AM–GM: $xyz = x + y + z \geq 3\sqrt[3]{xyz}$ gives $xyz \geq 3\sqrt{3}$, while $xy + yz + zx \geq 3(xyz)^{2/3} \geq 9 > 1$).
∎

Sanity: $x = y = z = \sqrt{3}$: sum $= 3\sqrt{3}$, product $= 3\sqrt{3}$ ✓, and $3\tan^{-1}\sqrt{3} = 3 \cdot \frac{\pi}{3} = \pi$ ✓.

WORKED EXAMPLE

Companion identity — $xy+yz+zx=1$ forces the sum to $\frac{\pi}{2}$

If $x,y>0$ and $xy<1$ then $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\frac{x+y}{1-xy}$; with $x,y,z>0$ and $xy+yz+zx=1$, pairwise: $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\frac{x+y}{1-xy}$ and $1-xy=z(x+y)$, so $\frac{x+y}{1-xy}=\frac{1}{z}$ and $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\frac{1}{z}=\frac{\pi}{2}-\tan^{-1}z$ (complementary pair!). Hence

$$\tan^{-1}x+\tan^{-1}y+\tan^{-1}z=\frac{\pi}{2}$$

$$\text{whenever } x,y,z>0,\ xy+yz+zx=1.$$

(Condition $xy<1$ is automatic: $xy<xy+yz+zx=1$ ✓.) Example: $x=y=\frac{1}{2}$, then $z=\frac{1-1/4}{1}=\frac{3}{4}$... check: $xy+yz+zx=\frac{1}{4}+\frac{3}{8}+\frac{3}{8}=1$ ✓ and $2\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{3}{4}\approx 0.9273+0.6435=1.5708=\frac{\pi}{2}$ ✓.

Nested and mixed constraint problems

STANDARD APPLICATION

Pattern 1 — nested arctans: $\tan^{-1}(\frac{1}{2})+\tan^{-1}(\frac{1}{5})+\tan^{-1}(\frac{1}{8})$-style chains

Add two at a time with the $xy<1$ face (small positive arguments keep you in the plain case): $\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{5}=\tan^{-1}\frac{7}{9}$ (since $\frac{1/2+1/5}{1-1/10}=\frac{7/10}{9/10}$); then $+\tan^{-1}\frac{1}{8}$: $\tan^{-1}\frac{7}{9}+\tan^{-1}\frac{1}{8}=\tan^{-1}\frac{65/72}{65/72}=\tan^{-1}1=\frac{\pi}{4}$. So the chain sums to $\frac{\pi}{4}$ exactly. The discipline: after each addition, re-check the xy condition with the new combined argument.

STANDARD APPLICATION

Pattern 2 — the branch constraints as a system: solve $\sin^{-1}x+\sin^{-1}y=\frac{\pi}{2}$ given $x+y=1$

Take the sine of both sides: $x\sqrt{1-y^2}+y\sqrt{1-x^2}=1$. With $y=1-x$ this becomes $x\sqrt{2x-x^2}+(1-x)\sqrt{2x-x^2}=\sqrt{2x-x^2}=1$, so $2x-x^2=1$, $(x-1)^2=0$, $x=1, y=0$. Check: $\sin^{-1}1+\sin^{-1}0=\frac{\pi}{2}$ ✓. Alternatively read the equation itself: $\sin^{-1}y=\frac{\pi}{2}-\sin^{-1}x=\cos^{-1}x=\sin^{-1}\sqrt{1-x^2}$ needs $y=\sqrt{1-x^2}$ (branch: $y\geq 0$) — together with $x+y=1$: $x+\sqrt{1-x^2}=1$ gives $x\in\{0,1\}$, and both $(1,0)$ and $(0,1)$ work by symmetry. ***Two solutions*** *— a squaring route may lose one; the branch route sees both.*

STANDARD APPLICATION

Pattern 3 — bounding an arctan sum without computing it

For positive a_i, $\tan^{-1}$ is increasing and concave on $\mathbb{R}^+$, but the exam-legitimate bound is cruder: $\tan^{-1} x < \frac{\pi}{2}$ and $\tan^{-1} x \le x$ for $x \ge 0$ (from the graph: tangent line at 0 is $y = x$, curve below it). So e.g. $\sum_{k=1}^{100} \tan^{-1} \frac{1}{k^2} < \sum_{k=1}^{100} \frac{1}{k^2} < \frac{\pi^2}{6} < 2$, and since also each term > 0 the sum is trapped in $(0, 2)$; sharper telescoping (oly-02) gives its exact value when the shape allows. Bounds + exact-when-possible is the olympiad habit.

COMMON TRAP

Traps — structure problems

- $x + y + z = xyz \Rightarrow$ sum $= \pi$ **requires positivity** (x, y, z of mixed signs can give 0 or $-\pi$; e.g. $x = y = z = 0$ satisfies the equation with sum 0).
- $xy + yz + zx = 1$ identity uses the complementary pair $\tan^{-1} \frac{1}{z} = \frac{\pi}{2} - \tan^{-1} z$ — true because $\frac{1}{z} > 0$; for $z < 0$ it fails (both sides off by π-jumps).
- In constraint systems, the range of each inverse is a real inequality: $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$ already forces $x, y \ge 0$ (LHS $\ge -\pi$, but for the sum to reach $\frac{\pi}{2}$ with both terms $\le \frac{\pi}{2}$ and $\ge -\frac{\pi}{2}$ you need... precisely: $x = \sin A, y = \sin B$ with $A + B = \frac{\pi}{2}$, $A, B \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ forces $A, B \in [0, \frac{\pi}{2}]$, so $x, y \ge 0$ and $y = \cos A = \sqrt{1 - x^2}$: the solution set is exactly the quarter circle $x^2 + y^2 = 1$, $x, y \ge 0$ — the constraint $x + y = 1$ then intersects it in $\{(1, 0), (0, 1)\}$ as computed).
- Two-step telescopes and nested arctans: re-check the addition-formula condition after **every** combination; a chain is only as safe as its last link.